REFERENCE: SPE-EUR-S-0009-TASI

DATE: 07/11/2025

ISSUE: 1 **Page:** 1/31

EUROPA

DC Harness Subsystem Requirements Specification

Written by	***Responsibility*** + handwritten signature if no electronic workflow tool
G. Capone	Harness Engineer
Verified by	
E. Pifferi	Head of Harness Engineering
R. Cialdini	Satellite System Engineering Manager
V. Iannucci	Product Assurance Manager
M. Prestianni	Configuration Manager
Approved by	
S. Palumbo	Program Manager

Approval evidence is kept within the documentation management system.

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CHANGE RECORDS

ISSUE	DATE	§ CHANGE RECORDS	AUTHOR
1	07-11-2025	First issue	G.Capone

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TABLE OF CONTENTS

1. INTRODUCTION 6

1.1 SCOPE AND PURPOSE 6

1.2 APPLICABLE DOCUMENTS 6

1.3 REFERENCE DOCUMENTS 6

2. REQUIREMENTS IDENTIFICATION 7

3. HARNESS GENERAL REQUIREMENTS 8

3.1 ELECTRICAL REQUIREMENTS 8

3.1.1 Interconnection wiring 8

3.1.2 Resistance 8

3.1.3 Insulation resistance and Dielectric strength 8

3.1.4 Wiring EMC classes 9

3.1.5 Lay wires priority 10

3.1.6 Twisted wires 10

3.1.7 Shielding of wires 10

3.1.8 Screen termination 18

3.1.9 Chain shield 18

3.1.10 Splice 19

3.1.11 Soldering 19

3.1.12 Solder screen termination and solder splice 19

3.1.13 Gold removal 19

3.1.14 Crimping of pin and socket 20

3.1.15 Crimping of splices 20

3.1.16 Insulation tubing application 20

3.1.17 Grommets 20

3.1.18 Harness Splices location 21

3.1.19 Over-shielding of bundles 21

3.1.20 Connectors protection 22

3.2 MECHANICAL REQUIREMENTS 22

3.2.1 Mass 22

3.2.2 Size 22

3.2.3 Harness Lacing 23

3.2.4 DC Harness Marking and Identification 24

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3.3 THERMAL REQUIREMENTS 25

4. ENVIRONMENTAL REQUIMENTS 26

4.1 TRANSPORTABILITY 26

4.2 MAINTAINABILITY 26

4.3 SAFETY 26

5. QUALITY ASSURANCE PROVISIONS 27

5.1 GENERAL 27

5.2 ACCEPTANCE TESTING 27

5.2.1 Harness resistance test 27

5.2.2 Harness dielectric strength and insulation test 27

5.3 CONNECTOR MATE / DE-MATE CONTROL 28

5.4 REWORK 28

5.5 MODIFICATION 28

5.6 HARNESS HANDLING 29

6. PREPARATION FOR DELIVERY 30

LIST OF FIGURES

Figure 3-1 Chain shields for D*MA 11

Figure 3-2 Chain shields for D*MA 12

Figure 3-3 Chain shields for D*MA 12

Figure 3-4 Chain shields for D*MA 13

Figure 3-5 Chain shields for D*MA 13

Figure 3-6 Chain shields for D*MA 14

Figure 3-7 Chain shields for D*MA 14

Figure 3-8 Chain shields for round connector 15

Figure 3-9 Chain shields for round connector 15

Figure 3-10 Chain shields for round connector 16

Figure 3-11 Chain shields for round connector 16

Figure 3-12 Chain shields for round connector 17

Figure 3-13 Chain shields for round connector 17

Figure 3-14 Harness crossing the S/C structure 18

Figure 3-15 Harness protections 21

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Figure 3-16 Harness tie-rap 23
Figure 3-17 Harness lacing 23
Figure 3-18 Connectors identification 24
Figure 3-19 Ty-rap and label 24

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1. INTRODUCTION

1.1 SCOPE AND PURPOSE

This document specifies the requirements applicable to the Manufacturing, Assembly, Integration and Test (MAIT) of DC Harness for EUROPA satellite and is applicable at S/C and P/L level.

1.2 APPLICABLE DOCUMENTS

Internal code / DRL	Reference	Issue	Title	Location of record
AD01	RQS-EUR-S-0003-TASI	1	S/C Electrical Design Requirements Specification (EDIS)	WAND
AD02	RQS-EUR-S-0014-TASI	1	PRODUCT ASSURANCE AND SAFETY REQUIREMENTS FOR SUB-CONTRACTORS	WAND
AD03	RQS-EUR-S-0004-TASI	1	S/C EMC/ESD Requirements Specification (EMC)	WAND
AD04	ECSS-E-ST-50-12C	1	SpaceWire – Links, nodes, routers and networks	
AD05	RQS-EUR-S-0012-TASI	1	S/C Thermal Design Requirements Specificification	WAND
AD06	RQS-EUR-S-0007-TASI	1	S/C Mechanical Environment Specification	WAND
AD07	RQS-EUR-S-0015-TASI	1	Space Radiation Environment Specification	WAND
AD08	PLN-EUR-S-0001-TASI	1	Configuration and Data Management Plan	WAND
AD09	TBD		Top Assembly Drawing	WAND
AD10	TBD		Mask Layout	WAND
AD11	ECSS-Q-ST-70-26C	1	Crimping of high-reliability electrical connections	
AD12	ECSS-Q-ST-70-08C	1	Manual soldering of high-reliability electrical connections	

1.3 REFERENCE DOCUMENTS

NA

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2. REQUIREMENTS IDENTIFICATION

The requirements specified herein are subject to revision by TAS-I when required.

In the event that a supplier believes that parts of the specified requirements would not apply, or additional requirements are necessary, such recommendations shall be submitted to TAS-I with substantiating analyses and documentation, for consideration and subsequent approval or rejection by TAS-I.

A unique identifier has been assigned to requirements in this document for traceability reasons. The requirement identifier is as follows:

- **PUID:** (Project Unique IDentifier):

e.g. EUR-XXXX-REQ-NNNNNN

where:

- EUR identifies the applicable project.
- XXXX identifies the Subsystem/Equipment/Unit and is composed of three to four digits, for this specification the acronym is DPH
- NNNNNN is the requirement number

Example of PUID for the “Ku/Ka Band Internal Calibration” Assembly:

EUR-RFKK-RQ-000120

- **Verification:** the minimum required verification methods are specified for each requirement in the field “Verification”, according to the following legend:
  - [T]: Test
  - [A]: Analysis
  - [R]: Review of Design
  - [I]: Inspection
  - [D]: Definition

The following language definitions are used to specify a requirement:

- **SHALL** is used to indicate a mandatory requirement.
- **SHOULD** indicates a preferred alternative but is not mandatory
- **MAY** indicates an option.
- **WILL** indicates a statement of fact or intention.

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3. HARNESS GENERAL REQUIREMENTS

3.1 ELECTRICAL REQUIREMENTS

3.1.1 Interconnection wiring

#EUR-DCH-000010 Interconnection wiring

Parent(s): none
Assignment:
Verification: [R, I]

The DC harness shall be manufactured according to the Harness Top Assembly Drawing as per [AD09].

##

3.1.2 Resistance

#EUR-DCH-000020 Harness continuity

Parent(s): none
Assignment:
Verification: [T]

The continuity of each conductor and of each shield shall be measured to verify the connection integrity.

##

3.1.3 Insulation resistance and Dielectric strength

#EUR-DCH-000030 Harness Insulation resistance and Dielectric strength

Parent(s): none
Assignment:
Verification: [T]

Each connection shall exhibit a resistance w.r.t. other connections and shields not less than 100 MΩ at test potential of 500 Vdc applied for 5 seconds without electrical breakdown or arc.

##

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3.1.4 Wiring EMC classes

#EUR-DCH-000040 Wiring EMC classes

Parent(s): EUR-EMC-REQ-000460

Assignment:

Verification: [R]

Wiring EMC classes are specified in [AD03], that specifies the harness classes 100,200,300,600 etc.

The following tables identify the cables typologies according to EMC classes and applied to the P/F.

HARNESS WIRING AND SHIELD			
Group	**Interface Types**	**Cable type**	**Remarks**
100	Main Bus Protected Lines	TP	
	Main Bus Direct Lines	TP	
	HET power lines	TP/TSP	Internal / External
	Main Bus Heater Lines	TP	
	Secondary power	TP / TSP	Internal / External
200	HV-HPC	TSP	
	HC-HPC	TSP	
	LV-HPC	TSP	
	ML / DS16 (RS422 I/F)	TSP / TWINAX (TCX)	125 Ohm/AWG 26
	1553 Data Bus	TWINAX (TCX)	75 Ohm/AWG24
	CAN Bus	TWINAX (TCX)	120 Ohm/AWG24
	Memory Load CMD (ML) via RS422	TSP	
	HPC1 commands	TSP	
	HL (14 V and 26 V) commands	TSP	
	Stretched Long Switch Closure	TSP	
	Serial Digital channels (DS-I6) via RS422	TSP	
	TC Spcf (synch, PPS)	TWINAX (TCX)	
	TC/TM video signals	TSP	
	Digital signals	TSP	
300	BSM	TSP	
	BDM	TSP	
	ASM (Single Ended)	TP	
	ASM (Double Ended)	TSP	
	Analogue signals	TP / TSP	
	TSM	TP	
400	NEA	TSP	
	Pyro	TSP	
	Thermal Knife	TSP	
500	RF signal	COAXIAL	50 Ohm
600	SPW	TSP	
	Space Fiber	COAXIAL/FIBER OPTIC	100 Ohm (COAXIAL)

Table 3.1.1.4 Harness wiring EMC classes

##

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3.1.5 Lay wires priority

#EUR-DCH-000050 Lay wires priority

Parent(s): none
Assignment:
Verification: [R, I]

- Wires shall be laid with the following priority:
    - 1° Lines of class 100
    - 2° Lines of class 300
    - 3° Lines of class 200/600
- For each class the following sequence shall be applied :
    - Single unshielded wires(S)
    - Twisted unshielded pair wires (TP)
    - Single shielded wires (SSH)
    - Twisted shielded pair wires(TSP)
    - Twinax wires and coax miniature cables

##

3.1.6 Twisted wires

#EUR-DCH-000060 Twisted Wires

Parent(s): EUR-EMC-REQ-000500
Assignment:
Verification: [R]

Twisting requirements shall be in accordance to [AD03].

##

3.1.7 Shielding of wires

#EUR-DCH-000070 Shielded wires

Parent(s): none
Assignment:
Verification: [R]

Shielded wires with at least 92% optical coverage shall be used. The maximum unshielded length of wires at the connector shall not exceed 25 mm.
The connection between shield and connector shall be shorter to minimize the inductance.
Shields termination shall be connected at both ends except the Thermistor lines (shield at thermistor shall be floating).

The connections (for d-sub connectors) will be realized by wires unshielded, soldered on the specific windows and connected around the backshell cable entry by metallic tie-rap according

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from Figure 3.1-1 to Figure 3.1-7
Shield Termination on circular connectors shall be made according to the mounting sequence described from Figure 3.1-8 to Figure 3.1-13.

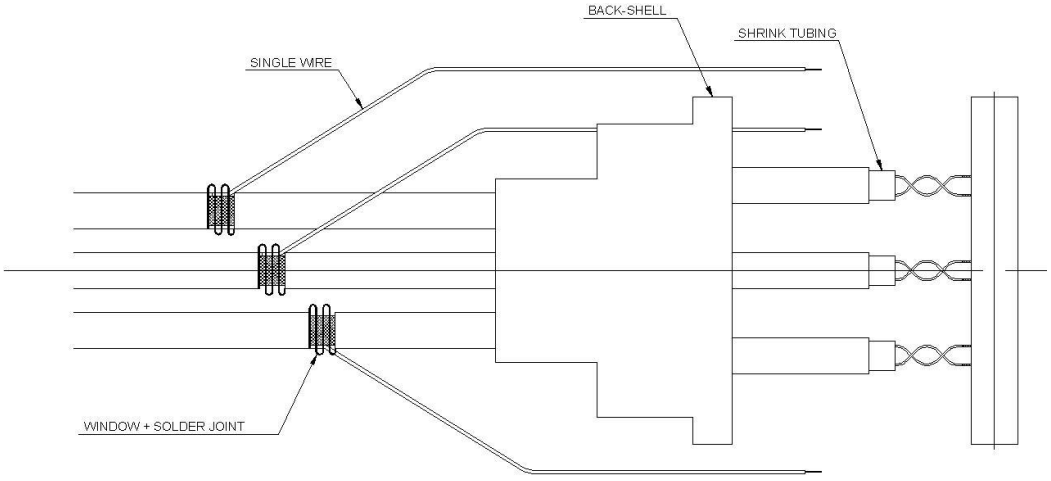


Figure 3-1 Chain shields for D*M

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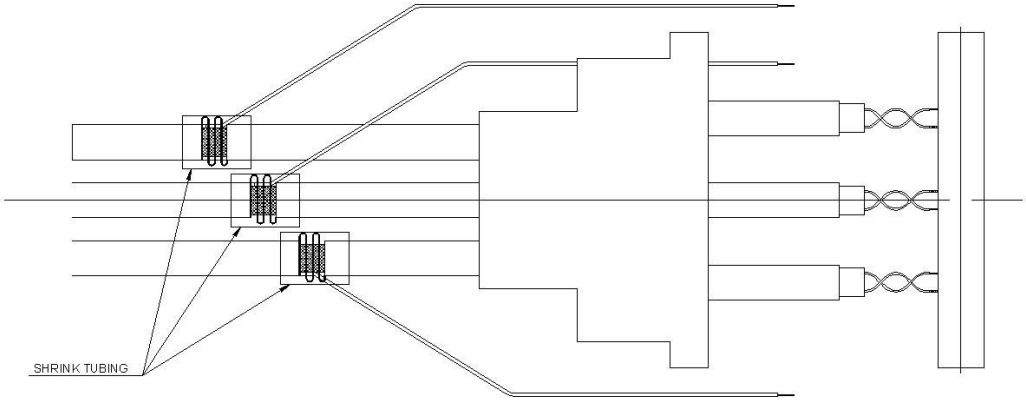


Figure 3-2 Chain shields for D*MA

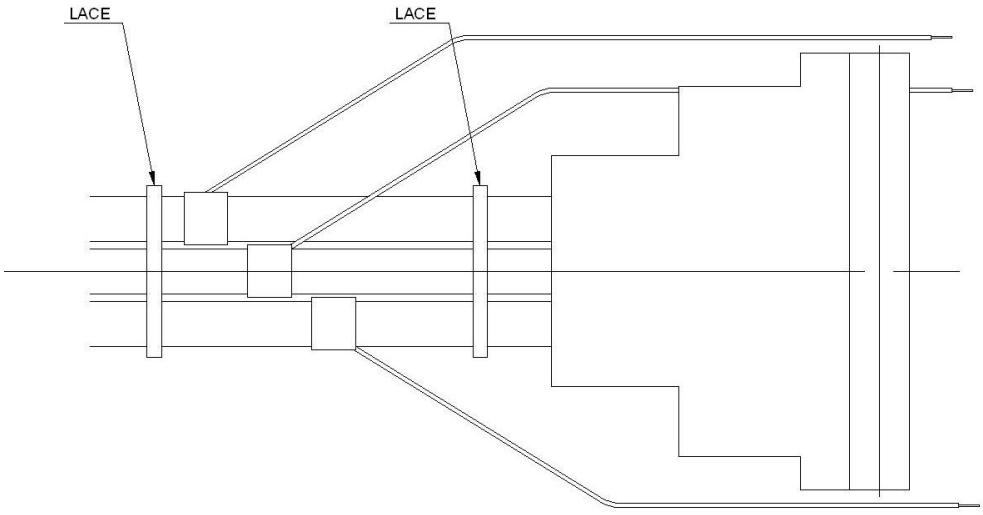


Figure 3-3 Chain shields for D*MA

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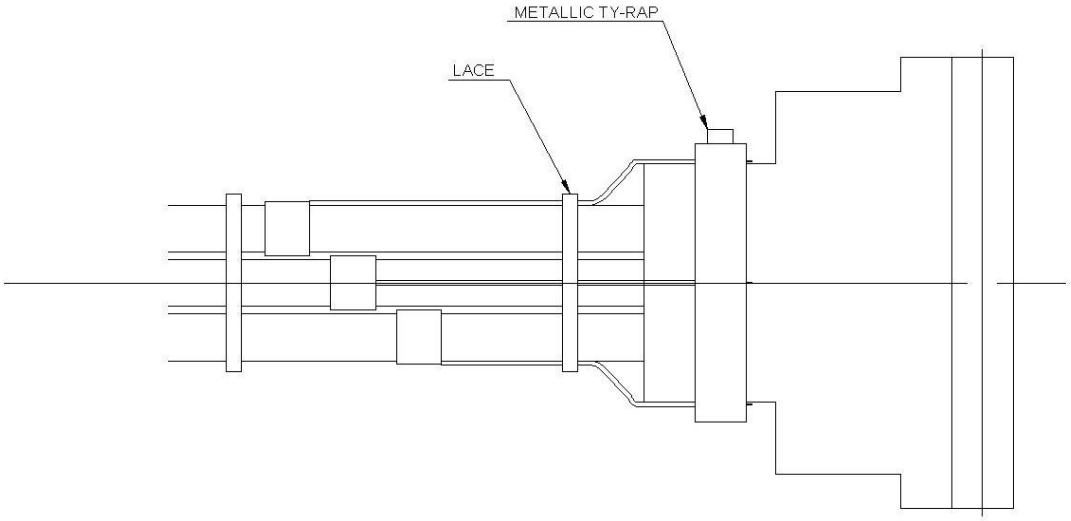


Figure 3-4 Chain shields for D*MA

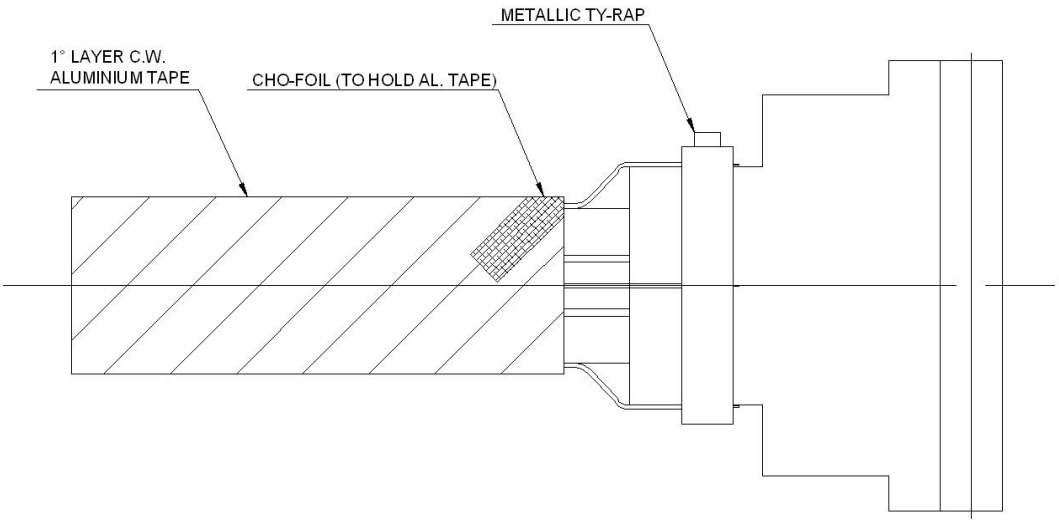


Figure 3-5 Chain shields for D*MA

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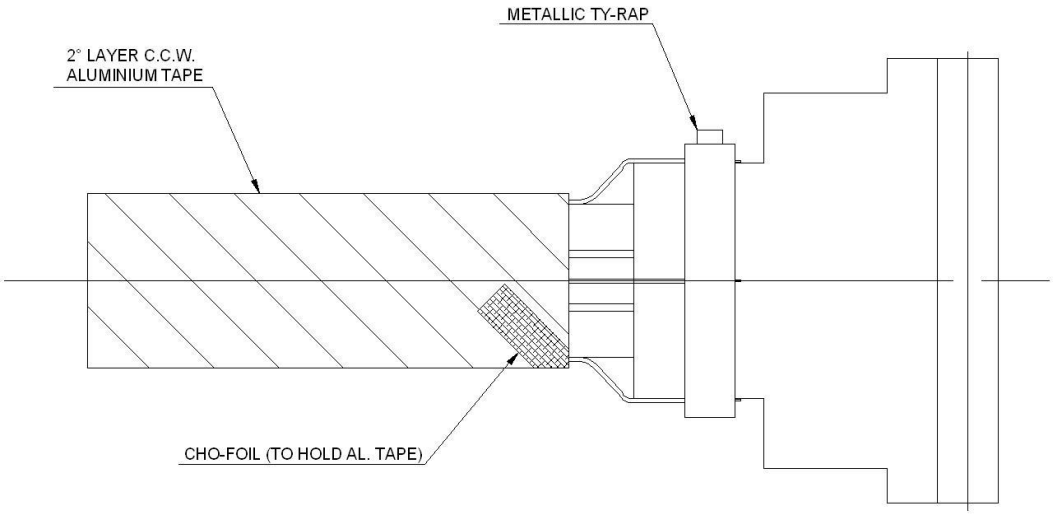


Figure 3-6 Chain shields for D*MA

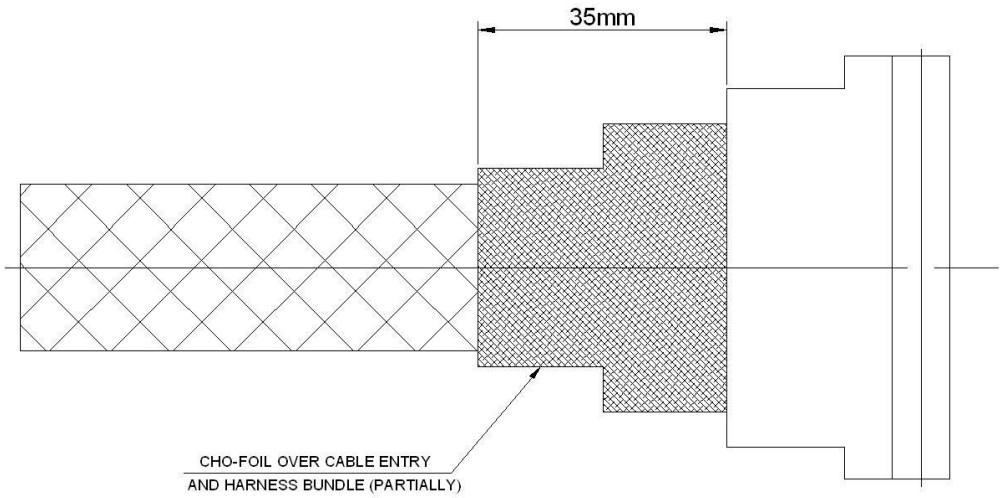


Figure 3-7 Chain shields for D*MA

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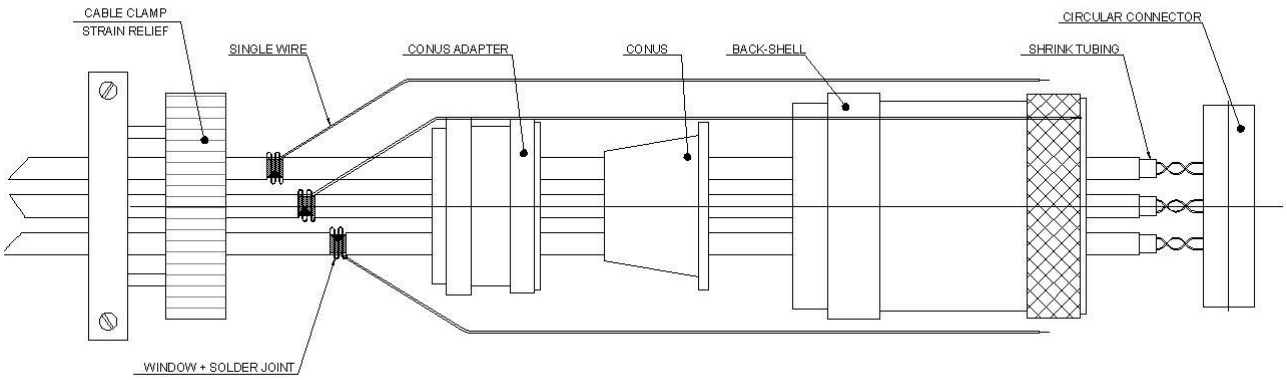


Figure 3-8 Chain shields for round connector

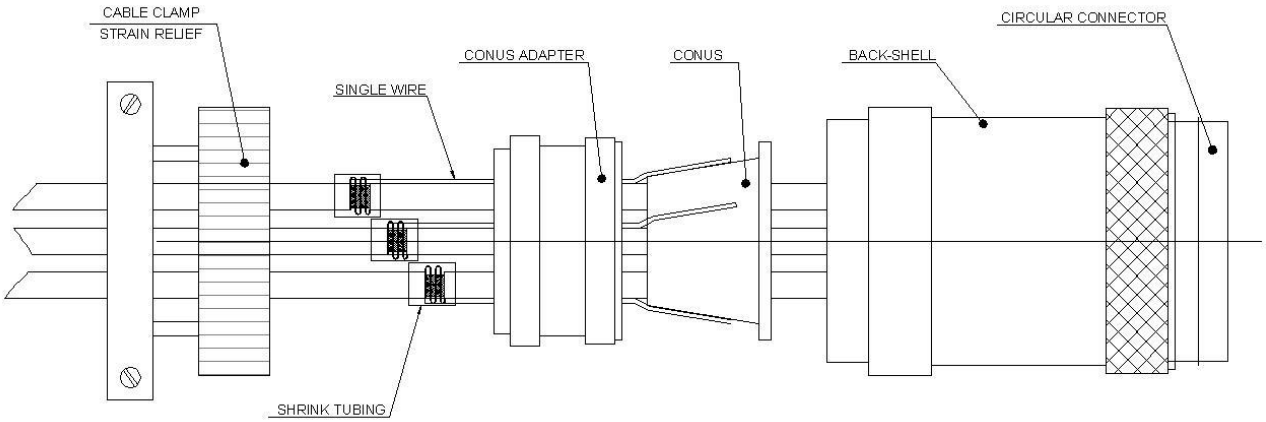


Figure 3-9 Chain shields for round connector

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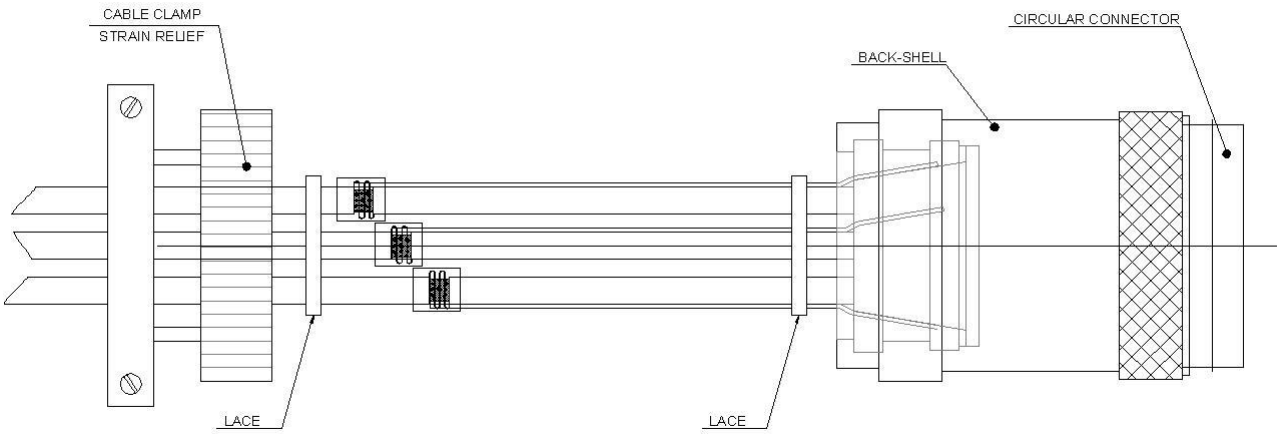


Figure 3-10 Chain shields for round connector

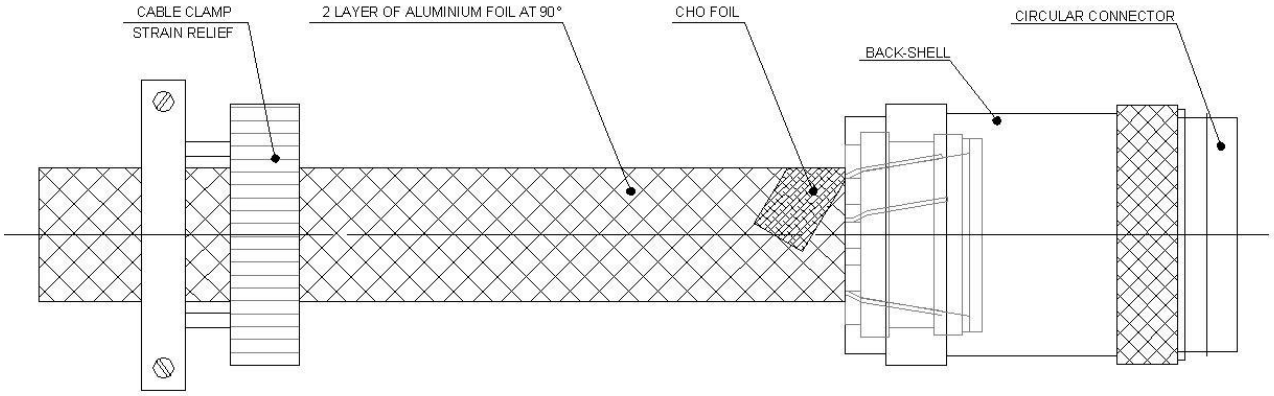


Figure 3-11 Chain shields for round connector

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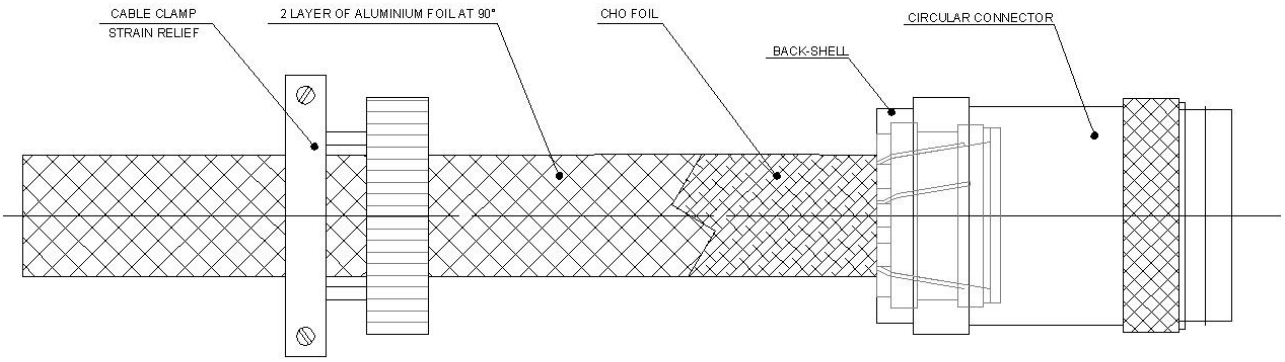


Figure 3-12 Chain shields for round connector

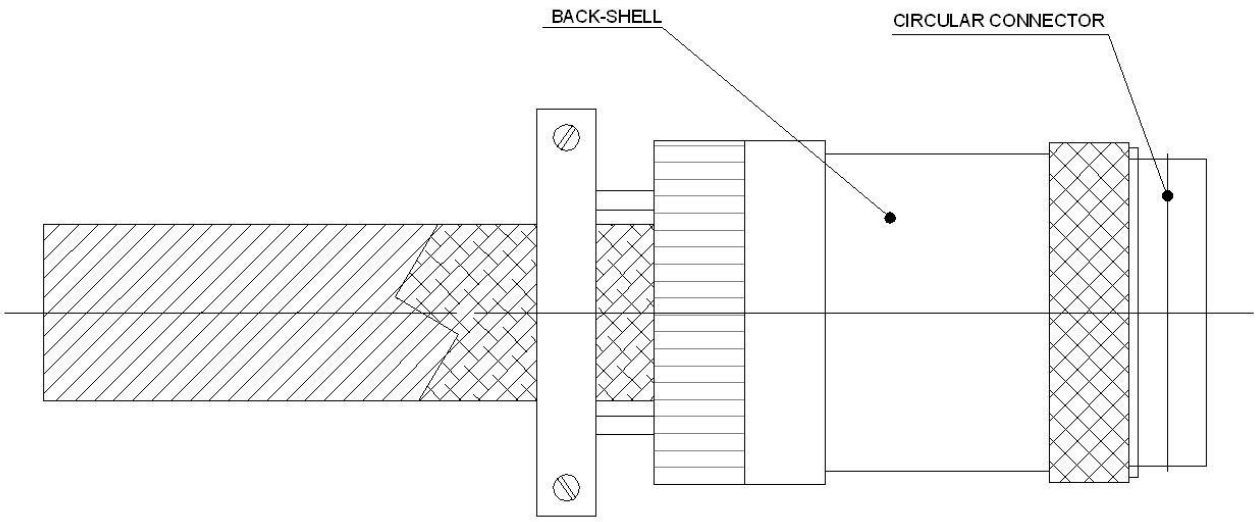


Figure 3-13 Chain shields for round connector

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#EUR-DCH-000071 Harness crossing the S/C structure
Parent(s): EUR-EMC-REQ-000575
Assignment:
Verification: [R, I]

If Harness routing foresees to pass from internal to external S/C structure, for EMI puropose it is mandatory to interrupt overall harness shield and to bond it separately to external and internal panel skin. All external bundles shall be covered not only with aluminium foil but also with a second shield made of braid shield.

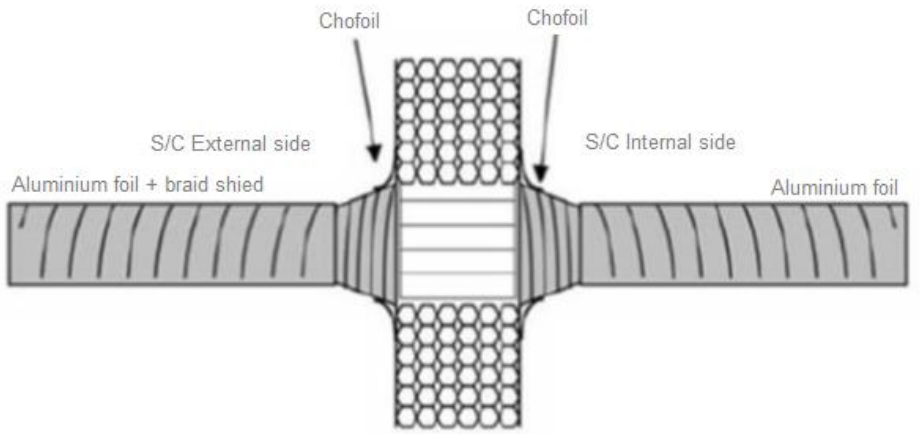


Figure 3-14 Harness crossing the S/C structure

3.1.8 Screen termination

#EUR-DCH-000080 Shielded wires screen
Parent(s): none
Assignment:
Verification: [R, I]

The shields of the wires shall be connected to the connector shell using flexible wires.
Shields terminated at circular connectors shall be electrically joined with no gaps around the full 360 degree circumference of the shield.
##

3.1.9 Chain shield

#EUR-DCH-000090 Harness chain shield
Parent(s): none
Assignment:
Verification: [R]

The daisy shields shall be forbidden.
##

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3.1.10 Splice

#EUR-DCH-000100 Harness splice

Parent(s): none
Assignment:
Verification: [R, I]

Splices shall be manufactured by crimping.

##

3.1.11 Soldering

#EUR-DCH-000110 Harness soldering

Parent(s): none
Assignment:
Verification: [R, I]

Soldered wire interconnections shall be made by a process which will allow for the removal of flux and flux residue, and in such a way as to permit full visual inspection of the interconnection and surrounding materials.

##

3.1.12 Solder screen termination and solder splice

#EUR-DCH-000120 Harness solder screen termination and solder splice

Parent(s): EUR-EMC-REQ-000310
Assignment:
Verification: [R, I]

Solder screen terminations and solder splices shall be made by a process in accordance to [AD03].

##

3.1.13 Gold removal

#EUR-DCH-000130 Harness gold removing

Parent(s): none
Assignment:
Verification: [R, I]

Gold shall be removed from components leads and terminal by dipping the leads into a solder bath, before soldering, in accordance to [AD12].

##

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3.1.14 Crimping of pin and socket

#EUR-DCH-000140 Harness crimping

Parent(s): none
Assignment:
Verification: [R, I]

All crimping shall be strictly controlled by means of detailed crimp schedules covering every combination of conductors screens and terminations qualified, calibrated tools, and test samples.
Crimping shall be in accordance to [AD11].

##

3.1.15 Crimping of splices

#EUR-DCH-000150 Harness crimping of splices

Parent(s): none
Assignment:
Verification: [R, I]

Crimping of splices shall be made in accordance to [AD11].

##

3.1.16 Insulation tubing application

#EUR-DCH-000160 Insulation tubing application

Parent(s): none
Assignment:
Verification: [R, I]

Space-qualified, heat-shrinkable insulation tubing shall be used for electrical insulation of solder splice, solder screen termination, crimping splice and on all the wires at the rear of the connectors (in this case RT850 is recommended).
For all soldering joints, the suitable heat-shrinkable tubing shall be selected (low recovery temperature will be selected, e.g. +100 C°), in order to avoid tin parts removal/damage, in accordance to [AD12].

##

3.1.17 Grommets

#EUR-DCH-000170 Harness grommets

Parent(s): none
Assignment:
Verification: [R, I]

Grommets or other protections shall be installed wherever harness is routed across protruding or sharp of structure, through or across unprotected holes or cut-outs

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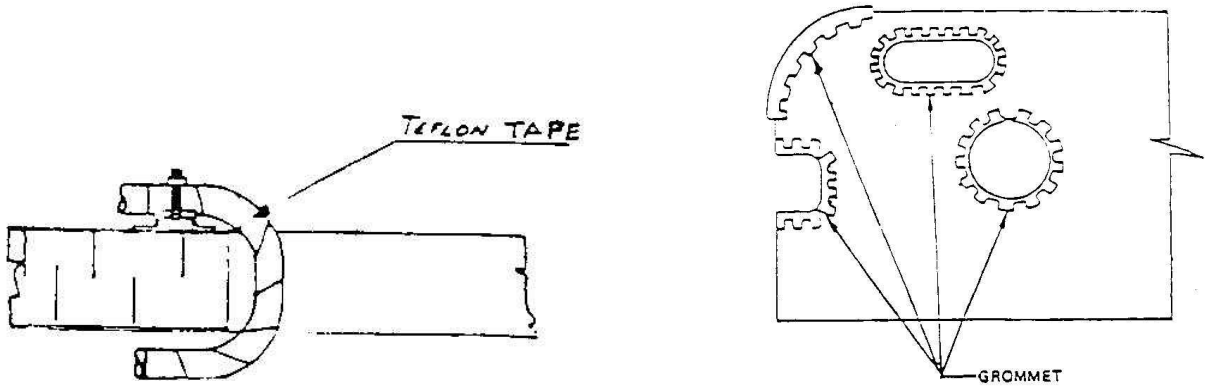


Figure 3-15 Harness protections

##

3.1.18 Harness Splices location

#EUR-DCH-000180 Harness Splices location

Parent(s): none
Assignment:
Verification: [R, I]

Splice shall be located inside the bundle secured to the structure, and in a part of the harness not subjected to stresses.

##

3.1.19 Over-shielding of bundles

#EUR-DCH-000190 Harness overall shielding and protection

Parent(s): none
Assignment:
Verification: [R, I]

The entire DC Harness shall be over-shielded and protected in accordance to [AD03].

##

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3.1.20 Connectors protection

#EUR-DCH-000200 Connectors protection

Parent(s): none
Assignment:
Verification: [R, I]

Metallic backshells shall be mounted on all harness connectors according to [AD03].

Limited number of thermal shrink backshell with conductive adhesive can be allowed upon Prime approval.

##

3.2 MECHANICAL REQUIREMENTS

3.2.1 Mass

#EUR-DCH-000210 Mass

Parent(s): EUR-SAT-REQ-005690
Assignment:
Verification: [T,R]

The Maximum DC & PYRO Harness mass including connectors, backshells and overall shields shall not exceed 110 Kg.

##

3.2.2 Size

#EUR-DCH-000220 Size

Parent(s): none
Assignment:
Verification: [R,I]

The DC harness manufacturing shall take into account the applicable layouts and routings as per [AD09] and [AD10].

##

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3.2.3 Harness Lacing

#EUR-DCH-000230 Harness lacing

Parent(s): none
Assignment:
Verification: [I]

The harness shall be secured by one of the following methods:

- Method A

  This method utilizes tie-rap mounted on the harness assembly and locked to prevent loosening or opening (Figure 3.2-1)

- Method B

  This method consists of not continuous lacing utilizing space lace (Figure 3.2-2).

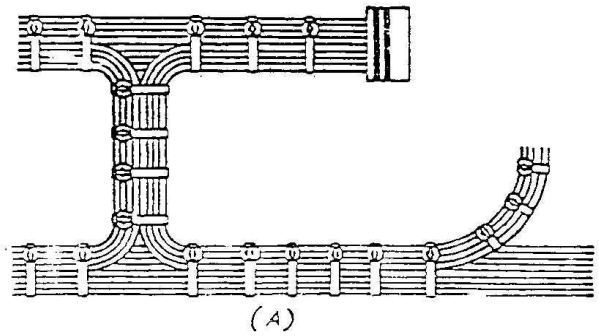


Figure 3-16 Harness tie-rap

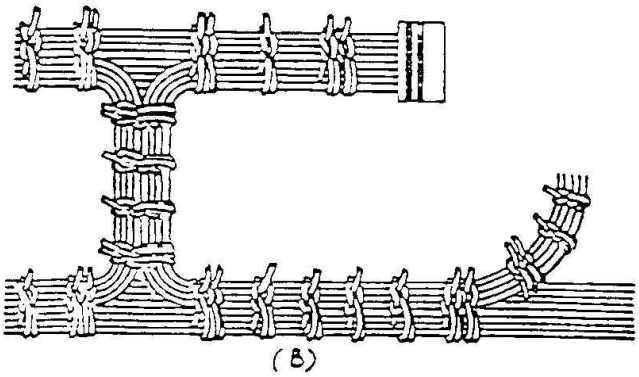


Figure 3-17 Harness lacing

##

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3.2.4 DC Harness Marking and Identification

#EUR-DCH-000240 Harness Marking

Parent(s): none
Assignment:
Verification: [R, I]

Each harness and harness connector shall have its reference number marked on thermos-shrink sleeve or identification plate to avoid mismatching (see Fig. 3.2-3, 3.2-4).

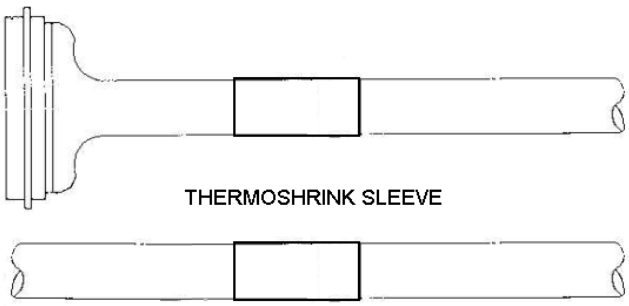


Figure 3-18 Connectors identification

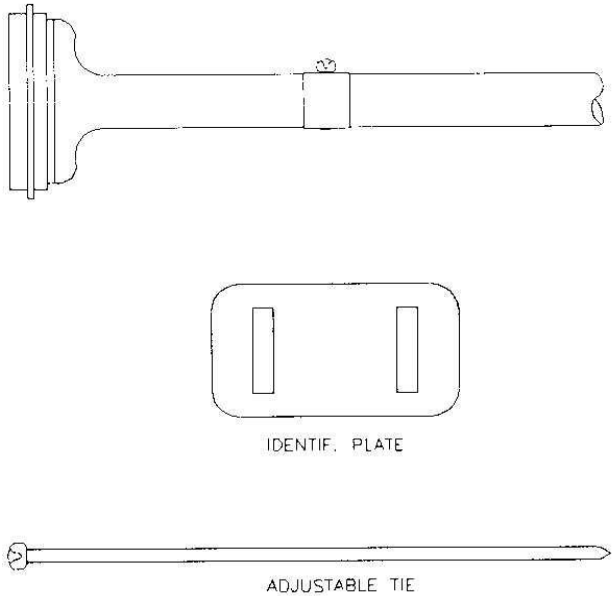


Figure 3-19 Ty-rap and label

##

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#EUR-DCH-000250 Harness Identification

Parent(s): none
Assignment:
Verification: [R, I]

DC Harness identification shall include at least the following information:

- Program
- CIDL
- P/N
- S/N

##

3.3 THERMAL REQUIREMENTS

#EUR-DCH-000260 Harness thermal requirements

Parent(s): none
Assignment:
Verification: [R, I]

The general interface requirements for thermal design of the harness are given in the applicable document [AD05].

##

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4. ENVIRONMENTAL REQUIMENTS

#EUR-DCH-000280 EMC conditions

Parent(s): none
Assignment:
Verification: [R, I]

The DC harness shall meet the specified performance requirements of [AD03],without degradation when exposed to the worst combination of specified operational environments.

##

4.1 TRANSPORTABILITY

#EUR-DCH-000290 Transportability

Parent(s): none
Assignment:
Verification: [R, I]

The DC Harness shall meet the transportability requirements given in [AD02].

##

4.2 MAINTAINABILITY

#EUR-DCH-000300 Maintainability

Parent(s): none
Assignment:
Verification: [R, I]

The DC harness shall be manufactured to be “maintenance free” in accordance to the requirements given in [AD02].

##

4.3 SAFETY

#EUR-DCH-000310 Safety

Parent(s): none
Assignment:
Verification: [R, I]

The DC harness shall be manufactured in order to be compliant with the safety requirements specified in [AD02].

##

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5. QUALITY ASSURANCE PROVISIONS

5.1 GENERAL

#EUR-DCH-000320 General

Parent(s): none
Assignment:
Verification: [I]

The inspection, tests and control of tests shall be conducted in accordance to [AD02].

##

5.2 ACCEPTANCE TESTING

#EUR-DCH-000330 Acceptance Testing

Parent(s): none
Assignment:
Verification: [T, R]

The harness, before delivery, shall be tested for continuity, resistance, dielectric strength and insulation.

##

5.2.1 Harness resistance test

#EUR-DCH-000350 Harness resistance test

Parent(s): none
Assignment:
Verification: [T, R]

The resistance test on power lines shall be performed by four-wires technique.

All values measured during the above test shall be stored and then reported into the test report.

##

5.2.2 Harness dielectric strength and insulation test

#EUR-DCH-000360 Harness dielectric strength and insulation test

Parent(s): none
Assignment:
Verification: [T, R]

There shall be minimum of 100 MΩ from each wire to all other wires of harness and from each wire to the back-shell when a potential of 500 Vdc is applied for a minimum period of 5 seconds, in accordance to [AD01].

##

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5.3 CONNECTOR MATE / DE-MATE CONTROL

#EUR-DCH-000370 Harness connector mate/de-mate control

Parent(s): none
Assignment:
Verification: [T, R]

Connectors shall be procured as Hi-Rel quality levels.

The following rules shall be applied during the harness manufacturing and test:

- Single pin or probe mating is forbidden.
- All mating shall be fully engaged.
- Flight connectors may only be mated a maximum of 20 times. Saver connectors shall be used to the maximum possible extent, to maintain the number of mate/demate operations as low as possible.
- Visual inspection shall be performed prior to each mate and after each de-mate in order to verify the connector correspondence and cleanliness.
- Any damage, contamination or excessive wear found during inspection shall be treated as a non-conformance.
- Connectors, when not mated, shall always be fitted with a clean protective cover; the harness shall be delivered with these covers in place.
- The number of mate/demate shall be reported in the harness log.
- Connectors savers shall be delivered with the harness connectors.

##

5.4 REWORK

#EUR-DCH-000380 Harness rework

Parent(s): none
Assignment:
Verification: [T, R]

When the harness requires rework after the final tests have been completed, the extent of the retest shall be according to NRB disposition, as per [AD02].

##

5.5 MODIFICATION

#EUR-DCH-000390 Harness modification

Parent(s): none
Assignment:
Verification: [T, R]

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Should the harness require any modification after it has been installed, the extent of the retest shall be in accordance with the NRB dispositions, as per [AD02].

##

5.6 HARNESS HANDLING

#EUR-DCH-000400 Harness handling

Parent(s): none
Assignment:
Verification: [I]

During harness manufacturing, safe precaution shall be taken to prevent:

- Abrasion or scratching
- Mechanical stress (torsion, traction, flexion, etc.)
- Improper vibration movement
- Dirtying

The wirings harness shall always remain secured to its harness fabrication mock-up, until immediately prior to being installed on the spacecraft.
During harness fabrication, handling and testing phases it is prescribed the use of the gloves and every precaution necessary for the cleanliness shall be taken according to [AD02].

The wiring harness fabrication and testing shall be performed in clean room (class 100000 temperature 22 C+/-2 C, 40 - 50 RH) according to [AD02].

##

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6. PREPARATION FOR DELIVERY

#EUR-DCH-000410 Preparation for Delivery

Parent(s): none
Assignment:
Verification: [R]

The end-item shall be handled, stored, packed, marked, labelled, transported / shipped in accordance to [AD02].

##

#EUR-DCH-000411 Shipping

Parent(s): none
Assignment:
Verification: [R, I]

The harness installed on satellite mock-up shall be transported in an opportune container.

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END OF DOCUMENT

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